

Vineel Koushik Peruri

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Education

S.R.K Institute of Technology, Vijayawada

2022 – 2026

B.Tech in Computer Science and Engineering (Data Science)

CGPA: 8.35

Technical Skills

Languages: Python, Java, JavaScript

Web Development: HTML, CSS, Node.js, Express.js, Flask

Data Science & ML: Pandas, NumPy, Scikit-learn, Matplotlib

AI/LLM Tools: LangChain, Pinecone, Hugging Face Transformers, Groq

Databases: MySQL

Core Concepts: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks

Experience

Data Science Intern, BIST Technologies

May 2025 – July 2025

- Developed a Road Accident Analysis project using real-world datasets to identify high-risk zones and accident patterns
- Performed data cleaning, preprocessing, and exploratory data analysis on large datasets
- Built machine learning models for classification and regression tasks to predict accident severity
- Visualized insights using Python libraries to support data-driven decision making

Projects

Medical AI Assistant (RAG Chatbot)

Tech: *Python, Flask, Pinecone, LangChain, Hugging Face Transformers, Groq*

- Developed a Retrieval-Augmented Generation (RAG)-based medical assistant to answer disease-related queries with improved contextual relevance
- Built a vector similarity search pipeline using embeddings, enhancing information retrieval efficiency
- Integrated 1000+ medical documents (PDFs and Wikipedia sources) to expand knowledge coverage
- Implemented query intent classification (definition, treatment, detailed response)
- Designed structured medical outputs including Symptoms, Causes, Treatment, Specialist, and Risk Level
- Created a responsive chat interface with real-time responses, improving overall user experience

Vendor Invoice Intelligence Dashboard

Tech: *Python, Pandas, NumPy, Streamlit, Matplotlib, Machine Learning*

- Built an interactive dashboard to analyze vendor invoice operations and financial data
- Automated detection of duplicate invoices and payment delays, reducing manual verification effort
- Applied data preprocessing, feature engineering, and machine learning models (classification & regression)
- Developed visualizations for KPIs such as vendor performance and payment cycles
- Delivered actionable insights to support data-driven financial decision-making

Problem Solving

- Solved 120+ problems on LeetCode, strengthening fundamentals in Data Structures and Algorithms
- Practiced core patterns including arrays, recursion, binary search, and dynamic programming
- Actively improving problem-solving skills with focus on time and space optimization